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Rubio pitches oil from U.S., Venezuela in Mumbai visit

External Affairs Minister hosts U.S. Secretary of State, UAE counterpart; Rubio expresses desire to send more Venezuelan oil to Ambani-owned plant in Jamnagar, praises India's refining capacities

Suhasini Haidar
NEW DELHI

Calling India a "refining superpower", U.S. Secretary of State Marco Rubio on Tuesday said he discussed possible Indian oil purchases from the U.S. during talks with External Affairs Minister S. Jaishankar, adding that India and the U.S. are "very, very close" to wrapping up negotiations on a much-delayed free trade agreement.

He will also discuss a plan for more Venezuelan oil to be refined at the Mukesh Ambani-owned Reliance Energy plant in Jamnagar, Mr. Rubio said.

Energy ties, trade, and connectivity were on the agenda as Mr. Jaishankar hosted Mr. Rubio and United Arab Emirates Foreign Minister Sheikh Abdullah Bin Zayed Al Nahyan in Mumbai. The External Affairs Minister met with each of his counterparts separately, and then held a meeting with both of them together, indicating that trilateral cooperation on the issues was also discussed.



External Affairs Minister S. Jaishankar with U.S. Secretary of State Marco Rubio and UAE FM Sheikh Abdullah bin Zayed Al Nahyan. ANI

Mr. Jaishankar called the trilateral meeting a "useful exchange of perspectives on energy, connectivity and contemporary developments".

The trilateral meeting, which was not earlier announced by the External Affairs Ministry, comes at a time when the UAE is facing many challenges in energy exports and trade due to the U.S. war in Iran and Tehran's blockade of the Strait of Hormuz. "The U.S., India, and the UAE have real opportunities to advance cooperation on energy, trade, technology, and regional connectivity," U.S. Ambassador to India Sergio Gor said in a social media post, sharing a photograph of the three leaders together in Mumbai.

However, the comments made no mention of plans for the 'I2U2' that the three countries formed with Israel in 2021 or the India-Middle East Economic Corridor (IMEC) that was announced in 2023.

CONTINUED ON
» PAGE 10
'PERFECTLY ALIGNED'
» PAGE 11

Terror ecosystem must be dismantled, Rajnath tells SCO

Saurabh Trivedi
NEW DELHI



Defence Minister Rajnath Singh at a meeting of the SCO Council of Heads of Government in Dushanbe, Tajikistan, on Tuesday. ANI

There can be no justification for terrorism and no double standard in tackling the menace, Defence Minister Rajnath Singh said on Tuesday as he called upon the member states of the Shanghai Cooperation Organisation (SCO) to dismantle the entire ecosystem of terrorism.

Speaking at the 25th meeting of the SCO Council of Heads of Government in Dushanbe, Tajikistan, in the presence of Pakistan's Deputy Prime Minister Ishaq Dar, Mr. Singh stressed that there can be no place for "those who provide safe havens to terrorists".

"Terrorism results in the loss of innocent lives and disrupts investment and commerce. It radicalises societies and deprives youth of the opportunities they rightfully deserve. There can be no justification for terrorism, no double standard in addressing it, and no place for those who finance, support, or provide safe havens to terrorists," Mr. Singh said.

He recalled Prime Minister Narendra Modi's remarks at the recent SCO Summit in Bishkek, Kyrgyzstan, calling for concrete steps on security, connectivity, and opportunity.

Mr. Singh said the bloc must adhere to the principles of sovereign equality, consensus, and mutual respect.

The Defence Minister also called for strengthening resilient supply chains, facilitating trade and investment, and leveraging technology to address global uncertainties arising from geopolitical tensions, trade disruptions, and concerns over energy and food security. India supported the adoption of an action plan for the multilateral trade and economic cooperation for the period 2026-2030, he said.

CONTINUED ON
» PAGE 10

China imposes restrictions on export of tunnel-boring, high-altitude machinery

Ananth Krishnan
BEIJING

China has said that it would introduce export controls, starting November 18, on tunnel-boring machines (TBMs) and other heavy machinery used in high-altitude terrain "to better safeguard national security and interests".

Announcing measures that will likely impact exports to India, China's Ministry of Commerce on Monday said "operators exporting the aforementioned items to sensitive countries and regions in dispute with China shall apply for a license", while exports to other countries will require prior registration to obtain certificates. The announcement listed TBMs and other heavy machinery, such as walking excavators and plateau-type wet shotcrete machines, used in construction in mountainous terrain and "altitudes above 3,000 metres". The Ministry said the items covered by the measures "have clear dual-use (civilian and military) characteristics".



The announcement comes amid India's concerns on clearance from China for imports of tunnel-boring machines. FILE PHOTO

"The measures are intended to better safeguard national security and interests, while also enabling China to better fulfil its international obligations, including those related to non-proliferation," the announcement said, adding that China had "notified the relevant countries and regions with which it has established export-control dialogue mechanisms."

The announcement comes days after Beijing unveiled a plan to impose sweeping export controls on 40 cutting-edge technologies including those for certain advanced solar manufacturing and some AI platforms.

India has previously raised with China the delays in customs clearance for imports of TBMs, which are used for several projects, including the construction of Metro rail networks and critical tunnelling projects in border areas.

CONTINUED ON
» PAGE 10

Unknown gaps

India must do better at detecting and preventing data breaches

The cybersecurity advisory by the Computer Emergency Response Team (CERT-In) has outlined two threat vectors *vis-à-vis* Indian commercial and government IT infrastructure: AI companies testing autonomous agents exceeding their designated operational constraints and bad actors weaponising open-source agentic AI tools for automated reconnaissance and exploitation. However, the organisations affected by recent breaches that the advisory was based on did not detect them: they came to light in the AI companies' disclosures and CERT-In operations. Aside from developers allowing experimental systems to act outside their authorised bounds, the advisory highlights two problems for India – existing arrangements focus on cybersecurity incidents after they are revealed; and the implicated organisations being found under-prepared for the DPDP Act – to be enforced from 2027, including steep penalties on entities not maintaining reasonable safeguards against breaches. While CERT-In observed or received reports of 29.4 lakh incidents in 2025, it does not routinely publish postmortem analyses and is exempt from the RTI Act. As a result, public accountability is also weak: CERT-In found that the AI agents and/or bad actors exploited vulnerabilities, including exposed passwords, that are known security problems separate from AI agents' abilities.

AI companies and labs have made several disclosures this year of agents unexpectedly breaching real-world infrastructure worldwide during testing. For India, however, the legal boundaries specified by the IT Act 2000 leave open an accountability gap for autonomous agents without the backing of human intentions. A more specific lacuna is that while CERT-In requires an organisation to report an intrusion within six hours, many organisations remain incapable of, or unwilling to, discover it. The reporting deadline itself has not ensured detection. Overall, the national cybersecurity landscape is fragmented across a pastiche of enforceable requirements in some domains, advisory technical standards in others, and narrow AI-specific accountability in yet others. As human and autonomous threat actors become more sophisticated, India needs to leapfrog to a landscape with enforceable risk-based requirements for both testing and defending against autonomous agents; better spotting unauthorised access, especially in the public sector; and a legal safe-harbour for security research conducted in good faith, in line with CERT-In policy. CERT-In should also publish more information, anonymised where necessary, which includes remediation outcomes and whether the affected individuals were informed. Finally, the government should also review the exemption for CERT-In from the RTI Act as such a sweeping exemption is unnecessary for effective cyber-defence.

Moves by the North of England to unlock India trade

The India-United Kingdom Comprehensive Economic and Trade Agreement (CETA) has opened the door to a new phase in the relationship between the two countries, with the focus now shifting from agreement to action. The momentum is already being seen, as the leaders from across the North of England have been travelling together to India, for the first time under The Great North banner to strengthen ties on trade, investment, innovation, culture and the visitor economy.



Kishore Jayaraman

The Group CEO of the UK India Business Council (UKIBC). He has been the India Chair of UKIBC and a Member of the UKIBC Group Board since 2016, and is also the former President of Rolls-Royce, India and South Asia

Beyond headlines

Agreements are only as powerful as the opportunities they unlock. Tariff reductions, regulatory alignment, and mobility provisions matter enormously; yet, their true value is realised only when businesses on the ground understand precisely where and how they apply. Keeping this in mind, the Northern Mayors were joined by businesses, small and medium enterprises and universities from across the region to help bolster links, showcase the North's economic assets and create new opportunities for trade, investment and academic collaboration. The North of England and India already have deep and growing ties, with nearly 5,40,000 people of Indian origin living across the region. Indian companies have invested over £1bn in the Northwest alone, supporting more than 6,700 jobs, according to the Confederation of Indian Industry report. Nearly 200 delegates from across the North were part of the visit, bringing together mayors, businesses, universities and leaders from tourism, culture and sport to make the case for the North as a destination for investment, partnerships and growth.

U.K. Foreign Secretary David Lammy's statement (made in 2024) describing the trade pact as "the floor, not the ceiling" of the bilateral relationship is the true essence of the current ties between New Delhi and London. The agreement is the result of several rounds of negotiations, over the years – 14 rounds of discussions, with each point deliberated to deliver and cash-in on strengths and individual goals of the two nations.

Regional partnerships can help businesses unlock the potential of the India-U.K. trade pact

The U.K. India Business Council, a mission partner for the Great North delegation, is working to help U.K. businesses better understand the opportunity and reframe how they see India. India is no longer simply a market to sell into. It is a strategic partner with the potential to become an integral part of the U.K.'s research and development (R and D), technology, talent and supply chains.

Visits to action

In the last few years, the ministerial and government visits have been largely with a broader vision to keep the ball rolling. This was seen with former U.K. Prime Minister Sir Keir Starmer's delegation coming to India in 2025 and vice-versa. But this time we see policymakers coming in to discuss the next steps. Government meetings, business delegations, university partnerships, R and D collaborations, sports and cultural ties and targeted regional engagements in business hubs of India. The itinerary was designed to build a long-term relationship.

There is already some impact, with the footprint of Indian businesses in the U.K. having expanded sharply this year, the number of Indian-owned companies rising nearly 60% from the previous year and combined revenues surpassing £105 billion, according to the latest India Meets Britain Tracker published by Grant Thornton UK.

Bilateral trade between India and the U.K. rose to £47.4 billion in 2025, marking an 11.7% increase from the previous year. Growth was concentrated on advanced manufacturing, clean energy and technology. With the CETA now in place, the India-U.K. corridor is poised for unprecedented growth, creating value for both economies.

In one of the UK India Business Council (UKIBC)'s reports on the CETA, it has focused on regional collaboration, mapping them against the sector-specific pathways for growth. A regional approach to trade engagement offers India a nuanced and targeted pathway into the U.K. economy; one that goes beyond London and into the innovation-rich heartlands of Britain. The report takes one step ahead from the macro view and delves into specifics. Examine Maharashtra's aerospace ambitions, Gujarat's chemical exports, Karnataka's world-class IT ecosystem and Tamil Nadu's pharmaceutical prowess. It asks not just what this deal means for India or the U.K. in aggregate, but also what it means for the industries and communities that will live with its consequences day to day.

Conversations need to translate into business. For a micro, small and medium enterprise, the challenge is often not whether the benefits of a trade agreement exist, but how one can access them. That means understanding and navigating the finer details, such as knowing that a tariff preference exists, establishing whether a product qualifies, meeting rules of origin, securing the right documentation, and managing the administrative costs of claiming the benefit. It can be seen as an empowering process.

Business guide

As India signs more trade agreements, the task is about ensuring businesses can take full advantage of what has been negotiated on their behalf. To simplify this, the UKIBC has developed a U.K.-India CETA Business Utilisation Manual – A Practical Activation Guide for Indian and UK Businesses, which was launched recently by Union Commerce and Industry Minister Piyush Goyal in London. The guide is designed to simplify the agreement and help businesses understand where the opportunities lie, whether they qualify, and how to act on them.

The ECI's unfinished lessons



Anurodh Lalit Jain

Vice-Chairman, AICC Minority Department. Views are personal

India's electoral system has evolved through successive efforts to make elections more inclusive, enforceable and accessible. Today, the mass revision of electoral rolls, suspicious voter deletions, and questions about access to polling records test the Election Commission of India's (ECI) ability to ensure that every eligible voter can participate equally. Its history offers lessons for the challenges ahead.

When India prepared its first electoral roll, nearly 28 lakh women were left off because they were identified through their relationships to men rather than being registered under their own names. Sukumar Sen, India's first Chief Election Commissioner, worked to ensure that women were enrolled in their own right, and by the 1957 elections, about 85% of those affected had been registered under their own names.

This lesson matters now as the ECI undertakes its Special Intensive Revision (SIR) of electoral rolls. In Bihar, around 65 lakh names were missing from the draft roll published in August 2025. The Commission cited reasons such as deaths, migration and duplicate entries. While updating electoral rolls is necessary, voters must be able to check why their names have been left out and challenge errors. The Supreme Court directed the ECI to publish the list of excluded voters along with the reasons for exclusion.

In a similar line, West Bengal shows what happens when correcting errors becomes slow. More than 27 lakh names were deleted during adjudication, prompting around 22 lakh appeals. Moreover, tribunal judges have raised technical concerns such as problems with the appeals portal, including difficulties issuing notices, conducting hearings and restoring names. While both cases show that removing names may be necessary to keep rolls accurate, correcting wrongful exclusions is equally important. A remedy that comes too late results in a lost vote.

While the ECI need not accept every allegation made by political parties, constitutional authority cannot substitute for verifiable evidence

The Form 7 question

The same principle applies to Form 7, through which an elector can seek the deletion of a name from the roll. In Aland, Karnataka, 6,018 online applications sought deletions. Of these, the ECI said that they rejected 5,994 applications as they were incorrect, and accepted only 24 after verification. An inquiry followed, and an FIR was registered. Rejecting incorrect applications show that safeguards can work. But a concentrated pattern of applications requires the ECI to establish who submitted them and whether identities or documents were misused. Every deletion request should leave a verifiable record of the applicant, supporting evidence, and the final decision.

Transparency in polling data presents another test. In Maharashtra's 2024 Assembly election, reported turnout rose from 58.22% at 5 p.m. to 66.05% in the final figure. The Commission explained that the earlier number was provisional and that data continued to be updated as polling parties returned and records were scrutinised. While a difference between provisional and final figures does not by itself establish improper voting, a substantial change makes a clear explanation necessary.

In December 2024, following a Punjab and Haryana High Court order to release election-related records in a case concerning the Haryana Assembly election, the government amended Rule 93 of the Conduct of Elections Rules. The change raised concerns about public access to electronic records, including CCTV footage and webcasting material. The timing of the change raised questions about whether the aim was to protect privacy or limit scrutiny.

Institutional independence matters just as much. In *Anoop Baranwal vs. Union of India* (2023), the Supreme Court prescribed an interim selection committee comprising the Prime Minister, the Leader of the Opposition and the Chief Justice of India to select election commissioners. Parliament later replaced the Chief Justice with a Union Cabinet Minister. The government now controls two of the three keys to the room where election commissioners are chosen. This hardly inspires public confidence in the ECI's independence.

While the ECI need not accept every allegation made by political parties, constitutional authority cannot substitute for verifiable evidence. Accurate rolls, traceable deletion requests, timely appeals and independently examinable polling records are safeguards owed to every voter. In an electoral democracy, trust must be earned through evidence, clarity and the willingness to correct mistakes.

Text&Context

THE HINDU

How conversion affects reservation

The invalidation of Tamil Nadu's order enabling persons converting to Islam from Backward Classes and Scheduled Castes to obtain Backward Class Muslim certificates highlights that while conversion cannot automatically transfer a person into a different notified community, continuing social disadvantage calls for an evidence-based inquiry

LETTER & SPIRIT

K. Kannan

The invalidation of Tamil Nadu's 2024 order highlights the distinction between Scheduled Caste status and backward-class classification. Conversion cannot automatically transfer a person into a different notified community, but continuing social disadvantage calls for evidence-based inquiry.



The Supreme Court recently upheld the Madras High Court's judgment invalidating Tamil Nadu Government Order dated March 9, 2024. SUSHIL KUMAR VERMA

THE GIST

- The Supreme Court upheld the Madras HC judgment invalidating Tamil Nadu's order enabling persons converting to Islam from Backward Classes and Scheduled Castes to obtain Backward Class Muslim certificates
- The State argued that a person who had enjoyed reservation should not lose every benefit by converting. The concern is understandable, but the order grouped persons from SC, BC, MBC and DNC classifications, without demonstrating a common social characteristic
- The larger question is whether the State should investigate evidence that converts continue to experience social exclusion and inherited disadvantage. Invalidating automatic transfer does not establish that fresh classification is impossible

Grounds for upholding HC ruling

On October 15, 2026, the Supreme Court, in *The Secretary to Government v. Sameer Ahamed N.*, upheld the Madras High Court's judgment invalidating Tamil Nadu Government Order dated March 9, 2024. The order sought to enable persons converting to Islam from Backward Classes (BC), Most Backward Classes (MBC), Denotified Communities (DNC) and Scheduled Castes (SC) to obtain Backward Class Muslim certificates and use the State's 3.5% reservation for BC Muslims.

The case arose from an application by a person born into a Hindu family who converted to Islam and sought a certificate as a Muslim Lebbai. The High Court held that conversion did not make the applicant a member of any of Tamil Nadu's seven notified backward Muslim communities: Ansar, Dekkani Muslims, Dubekula, Labbais (including Rowthar and Marakayar), Mapilla, Sheik and Syed. The Supreme Court affirmed that conclusion. The courts rejected the assumption that a former caste classification establishes membership of a different Muslim community. A person does not become a Lebbai, Rowther or Dekkani Muslim merely by converting. A Jamath may attest to conversion but cannot conclusively determine eligibility for a statutory reservation category. A community certificate records legal status; it cannot create it.

The State argued that a person who had enjoyed reservation should not lose every reservation benefit by converting. The concern is understandable, but the method was legally unsustainable. The order grouped persons from SC, BC, MBC and DNC classifications, without demonstrating a common social characteristic linking them to the seven notified Muslim communities. The verdict rejects automatic transfer of status; it does not, however, declare that Muslims can never qualify as a backward class.

Two constitutional routes

The Constitution treats Scheduled Caste status and backward class identification differently. Article 341 empowers the President to specify Scheduled Castes; Article 341(2) provides that Parliament may alter the list by law. Paragraph 3 of the Constitution (Scheduled Castes) Order, 1950 restricts SC status to persons professing Hinduism, Sikhism or Buddhism. Under the existing Order, conversion to Islam or Christianity does not preserve SC status merely because of former caste identity.

Backward-class identification follows another route. Articles 15(4) and 16(4) permit special provisions for socially and educationally backward classes and for their adequate representation in public employment. Article 342A, as amended by the Constitution (One Hundred and Fifth Amendment) Act, 2021, recognises States' power to maintain their own lists of socially and educationally backward classes for State purposes.

Backward classes need not be identified exclusively through Hindu caste names. Occupational groups, sects and communities among non-Hindus may qualify if constitutional criteria are met. But the State cannot establish qualification merely by assigning converts to an existing Muslim community without evidence of membership or a defensible basis for a fresh classification.

The Mandal Commission's indicators

The Second Backward Classes Commission, chaired by B.P. Mandal, offers a useful comparison. It used 11 indicators across social, educational and economic criteria. Social indicators carried three points each, educational indicators two, and economic indicators one, for a maximum score of 22. Its aim was to identify communities experiencing combined disadvantage, not simply reproduce caste names. The indicators need not be mechanically applied to every later classification. The enduring principle is that identification requires a reasoned assessment of disadvantage.

Caste has often been the principal organising unit of reservation. Once a caste is recognised as backward, an individual ordinarily proves membership rather than personal backwardness afresh. Yet the constitutional purpose is to advance socially and educationally backward classes, not preserve labels regardless of present conditions.

A person from a Hindu MBC may face inherited exclusion after conversion to Islam. But the former caste name does not establish membership of a Muslim community already notified as backward. Equally, the absence of caste hierarchy as a religious principle in Islam cannot prove, without inquiry, that inherited disadvantage has disappeared. The State should identify a recognisable group among converts, assess its continuing social and educational position, and explain why the evidence warrants inclusion in its backward-class list.

T.N. Commission's views

The Government Order followed a letter dated February 6, 2024, from the Chairman of the Tamil Nadu Backward Classes Commission. The letter of the Commission actually records the fact that one of the members had observed that for extension of benefit of consideration for a backward class Hindu to Islam, quantifiable data may be collected by the Government to show how many such people converted to Islam have availed the benefits. This was not yet good enough. The State is reported to have said that it had considered the Commission's views, following stakeholder engagement and deliberation. Neither the Commission nor the State conducted any empirical inquiry establishing that converts from different Hindu classifications shared a relevant characteristic with the seven notified Muslim communities.

As described in the judgment, the recommendation proposed rationalising BC Muslim certificates for converts from specified Hindu classifications. It did not establish the link between a person's former classification and membership of a particular Muslim community.

A Hindu BC, an MBC member, a member of a Denotified Community and a person recognised as belonging to a Scheduled Caste do not necessarily experience the same disadvantage. Grouping them together cannot be justified solely because conversion may affect their previous reservation entitlement. The question is not whether the Government acted with benevolent intent, but whether its means were constitutionally permissible.

The Sachar Committee

The Sachar Committee, constituted in 2005 under Justice Rajindar Sachar, examined Muslims' social, economic and educational status. Its 2006 report documented disadvantage in education, employment and access to credit.

The report shows why social position must be examined through evidence rather than inferred from religious identity. It did not establish that Hindu converts from particular BC or MBC communities formed a distinct backward class under Tamil Nadu's framework. Nor does evidence of general disadvantage among Muslims prove membership of one of the seven notified communities.

Religious identity, caste history, occupation, location, educational attainment and representation in public employment may all be relevant. Their relationship, however, must be demonstrated rather than presumed.

Distinct legal inquiries

The National Commission for Religious and Linguistic Minorities, chaired by Justice Ranganath Misra, reported in 2007 and recommended delinking SC status from religion by deleting paragraph 3 of the 1950 Order. It argued that the treatment of historically disadvantaged castes should not depend on the religion their members profess. The recommendation did not amend the Order; its restriction remains unless changed through the prescribed constitutional process.

The Commission led by former Chief Justice of India K.G. Balakrishnan, constituted in October 2022, was tasked with examining whether SC status could be accorded to persons claiming historical SC origins who converted to religions outside those specified in the Presidential Orders. Its remit concerns the consequences of caste-based disadvantage after conversion.

This is distinct from the Tamil Nadu BC Muslim issue. The SC question arises under Article 341 and the 1950 Order; backward-class identification arises under Articles 15(4) and 16(4), and requires identification of a socially and educationally backward class under the State framework. The Mandal, Sachar, Ranganath Misra and Balakrishnan exercises illuminate different questions and cannot substitute for one another.

A way forward

The ruling means that conversion to Islam does not automatically carry forward a former Hindu BC, MBC or DNC classification into BC Muslim status, or permit assignment to a notified Muslim community without a lawful basis for membership. The separate restriction governing SC status under the 1950 Order remains unaffected.

The larger question is whether the State should investigate evidence that converts continue to experience social exclusion, inherited disadvantage or educational deprivation. Invalidating automatic transfer does not establish that fresh classification is impossible. Tamil Nadu could commission an independent study through its Backward Classes Commission, examining educational attainment, representation in public employment, social acceptance, and the experiences of converts from different BC, MBC and Denotified Communities.

The inquiry should not treat all converts as homogeneous. The State must define any proposed class, explain its relationship with existing classifications and demonstrate that it meets constitutional criteria.

(K. Kannan is a Senior Counsel and Mediator, a former judge of the Punjab and Haryana High Court and Chairman of the Railway Claims Tribunal.)

CACHE



Growing market: People wait for their electric scooters at a vehicle charging and battery-swapping station in Bengaluru. SUDHAKARA JAIN

Can India turn retrofitment into the next mobility revolution?

Unlike new EV manufacturing, which rewards scale and centralisation, retrofitment rewards distribution through local workshops, technicians and entrepreneurship; with the right training, neighbourhood mechanic shops can become certified retrofitters, absorbing the existing workforce

Jaideep Saraswat
Ashish Dokania

For decades, India's growth story has been written in a familiar script: take a technology invented elsewhere, strip it down, re-engineer it for cost, and scale it for the masses. Generic drugs, low-cost telecom, frugal automobiles are examples of this approach. It is a script that India has performed brilliantly. The question India must ask now is whether it is ready to move from optimisation to creation.

There are signs that it already has. Digital payments existed long before Unified Payments Interface (UPI), yet UPI reimagined the entire architecture of a transaction and created a model good enough that other countries want to emulate. Retail existed long before quick commerce, yet ten-minute delivery rewired consumer expectations in a way no global template had. In both cases, India moved beyond the familiar script and engineered a new category of implementation for existing systems. These models are suited to India's density, informality, and scale. Now, they are travelling across borders. Mobility presents a similar opening.

Vehicle retrofitment

Vehicle retrofitment sits at the centre of the new opportunity. Retrofitment, which converts an existing petrol or diesel vehicle into an electric one by replacing the powertrain, keeps the chassis intact. It is a useful technical fix, but a genuine innovation opportunity arises from everything that needs to be built around it – from diagnostics for legacy vehicles, battery and power-electronics systems designed for retrofitted architectures, calibration software, testing protocols, safety certification, financing solutions, installation networks, and a skilled service workforce. Innovation arises from moving beyond a single product to creating an entire operating system for a new industry, whose definitive version is not built yet.

India has the raw ingredients to attempt it: a deep automotive supply chain, engineering talent that has already proven itself in frugal design, a fast-growing EV sector, credible software capabilities, and critically, a domestic market large and varied enough to pressure-test solutions.

The decentralisation edge

India's advantage is structural. Unlike new EV manufacturing, which rewards scale and centralisation, retrofitment rewards distribution like local workshops, local technicians, and local entrepreneurship. India has already shown what decentralised infrastructure delivery looks like at scale.

The PM Surya Ghar Yojna (PMSGY) on rooftop solar has now reached over 63 lakh households, built through a network of more than 34,000 empanelled vendors that has generated more than 4.6 lakh jobs across the solar rooftop value chain. Here is the blueprint for what a retrofitment ecosystem could look like, encompassing not a few large factories, but thousands of certified local centres. Each centre can convert, service, and maintain vehicles while generating an average of at least 20 direct and indirect jobs.

India's automotive geography is not confined to its metros. Millions of vehicles run in Tier-II and Tier-III towns and rural India, held together by an enormous, largely informal network of mechanics, garages, and spare-parts dealers. With the right standards and training, that existing workforce can be absorbed, converting neighbourhood mechanic shops into certified retrofitters. That proximity between service providers and vehicle owners also solves a trust problem that new technologies always face.

An opportunity for the Global South

The opportunity extends beyond India's border. Countries across South Asia, Southeast Asia, Africa, and Latin America share the same three conditions that make retrofitment relevant in India. They have large fleets of ageing vehicles, price-sensitive consumers, and ownership cycles far longer compared to wealthier markets. Collectively, the Global South spends more than $550 billion a year importing oil, a dependence that new EV sales alone will take decades to meaningfully dent, since new vehicles replace only a fraction of the fleet each year.

Retrofitment offers these countries a faster and cheaper route to displacing that fuel bill using vehicles already on their roads.

This is precisely the shape of opportunity UPI has already proven India can capture. UPI-linked payments are now live in eight countries, and India has signed digital-infrastructure cooperation agreements with 23 more. This was realised not because India sold a payments app, but because it exported a replicable model and an implementation playbook that other countries could adapt to.

Retrofitment could follow the same arc. If India builds the certification standards, financing models, safety protocols, and training pipelines for retrofitment near home, and at scale, it will have effectively built an exportable system.

Other nations with similar vehicle fleets and fuel-import pressures would rally behind it.

None of this happens by accident. It requires India to treat retrofitment, not as a stopgap measure to be tolerated until new EVs become universally affordable, but as a genuine innovation category to be engineered deliberately. India has the right pieces, the engineering talent, the manufacturing base, the technician networks, and the domestic testing ground. Now it needs the intent to bring these pieces together for retrofitment at the scale of PMSGY.

(Jaideep Saraswat is the Associate Director of the Clean Power, Electric Mobility & Emerging Technologies vertical at Vasudha Foundation. Ashish Dokania is the CEO of Green Tiger Mobility Private Limited. The authors declare no competing interest.)

From Page One

Rubio pitches U.S., Venezuela oil to India

The U.S. official's visit comes at a time of tension in India-U.S. ties due to a number of issues, including the U.S.'s recent moves to stop H-1B visas, threatened sanctions (that have since been held off) over India's import of Russian oil, and delayed FTA negotiations.

"We're very, very close. I would say we're about as close as you can be without having executed the [free trade] agreement. I think there's a small number of items left to wrap up, so I think that will happen shortly," Mr. Rubio said, adding that the State Department was involved closely with the negotiations between trade officials.

While Mr. Jaishankar met Mr. Abdullah in Mumbai just four days after he had travelled to Abu Dhabi where the two Foreign Ministers had also spent time together, this is Mr. Rubio's second visit to India in five months after his last visit to Delhi for the Quad Foreign Ministers meeting in May this year. This time around, Mr. Rubio's focus is on the economic side of the bilateral relationship, officials said.

There have been questions raised about Mr. Rubio's planned two-day visit to Jamnagar, as well as criticism that he had "summoned" Mr. Jaishankar to Mumbai rather than travelling to meet him in Delhi. Speaking to journalists at the Taj Mahal Palace Hotel in Mumbai where he and Mr. Jaishankar offered tributes to those who were killed in the 26/11 terror attacks, Mr. Rubio fielded questions over the purpose of his visit, and why he had returned to India so soon.

"First of all, I will point out that the Foreign Minister of India has been to the United States four times this year," Mr. Rubio explained, adding that as the most populous country and largest democracy that is a trade and defence partner, a "strong strategic partner" and a Quad member, India is an important country to interact with.

Reviving Reliance's Venezuelan link

He explained that his visit to Jamnagar, where he will see the Reliance energy refinery and the Vantara animal facility, was in order to discuss more Venezuelan oil being brought to the refinery.

"We've talked to them [Reliance Energy] about it in the past. We're going to go see what the holdup is and what opportunities exist. We'll see how the conversation goes. This is a company that has had a presence in Venezuela in the past and whose refining capabilities are substantial," he added.

When asked about India's purchase of Russian oil that makes up about 45% to 50% of its intake, Mr. Rubio said that he was "happy" to see India has made "substantial progress" on diversifying its oil purchases, adding that the U.S. and Venezuela can be the source of more energy inputs to India.

(with inputs from Vinaya Deshpande in Mumbai)

Terror ecosystem must be dismantled: Rajnath

New SCO mechanisms should address clearly identified needs and avoid duplication, Mr. Singh said. He welcomed the decisions to establish the Universal Centre in Tashkent and the Anti-Drug Centre in Dushanbe as permanent SCO bodies, while stressing the need for effective coordination and timely implementation.

Thorny connectivity projects

Highlighting regional connectivity, Mr. Singh noted that initiatives such as the International North-South Transport Corridor and the Chabahar Port in Iran could connect Central Asia with wider markets. However, connectivity projects must respect the sovereignty and territorial integrity of all countries, he said.

The meeting reviewed economic and trade cooperation among member states and discussed transport, finance, the digital economy, and social development and the Heads of Delegation adopted a joint communiqué, the SCO budget for 2027.

Mr. Singh also invited member states to participate in the first SCO Civilisational Dialogue Forum, to be hosted by India later this year.

China imposes curbs on machinery exports

On Monday, China also announced export control measures related to certain types of drones. The controlled items include "autopilots specially designed for Unmanned Aerial Vehicles (UAVs); UAVs or unmanned airships with a maximum endurance of three hours or more; certain engines specially designed for UAVs or unmanned airships; and UAVs with a range of 300 kilometres or more" among other items.

The latest announcements followed Beijing on October 16 unveiling a sweeping draft plan to restrict exports of more than 40 cutting-edge technologies, including solar manufacturing technologies, some AI models, and certain biotech platforms.

A number of technologies related to the manufacture of solar cells have been listed for either a complete export ban or restricted exports. While the draft plan does not restrict most solar exports, the new controls on next-generation solar manufacturing technologies could impact efforts by countries including India to reduce dependence on China as they look to move up the solar manufacturing chain.

Report flags feature to centrally control access on ECINET

Sreeparna Chakrabarty
NEW DELHI

The ECINET, the Election Commission's comprehensive digital platform, has a feature through which viewing of enumeration form details is enabled State by State. This means that access for State Nodal Officers (SNOs) could be switched off anytime making it a centralised platform, according to an audit by 'Charchagram', a civic tech initiative.

The report assumes significance in view of objections raised by Election Commissioners Sukhbir Singh Sandhu and Vivek Joshi over the centralisation of ECINET and the electoral roll database.

The ECINET has a dedicated State Nodal Officer management area (create, manage, update) with a built-in disable action, according to the report, authored by Saurav Rajput, a data scientist and an alumnus of IIT, Delhi and IIM, Calcutta.

Enable, disable buttons

As far as State-level access was concerned, ECINET has an SNO module behind the official login. Each SNO row has a disable button and the disable pop-up says: "On disabling this State Nodal Officer, the user will no longer be able to access the application."

Separately, the code has a message: "View enumeration form details is not enabled for selected State." This, in effect, means that viewing the enumeration form can be switched on or off State by State which, in turn, means that access for State Nodal Officers (SNOs) can be switched off.

Access for State Nodal Officers to view enumeration forms could be switched off, it says

The feature can be switched on and off per State, and for States where it is not enabled, officials get a message "View Enumeration Form Details is not enabled for selected State" instead of the form details.

While it was not clear since when this feature was available on ECINET, it might be recalled that the portal itself was launched in January 2026. The ECINET is the ECI's unified portal for officials. Mr. Rajput said he could locate the feature since September 28. The audit says ECINET also carries a feature that can bar electoral registration officers (EROs) and assistant EROs (AEROs) only in West Bengal from reviewing voters' enumeration forms and viewing the notice-and-hearing dashboard.

The Congress said: "It is by now well-established how [CEC] Gyanesh Kumar, at the instance and instructions of the PM (Narendra Modi) and the HM (Amit Shah), put in place ECINET to centralise all manipulations of the election process."

"The law establishes that EROs have statutory authority over the electoral roll. But across the country, the ECINET system and the Election Commission's IT division have overruled and violated these safeguards," Congress leader Jairam Ramesh said in a post on X.

Aditya-L1 mission offers new insights into the direction of solar eruptions

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BENGALURU

Scientists have generated Doppler velocity images of activities taking place in the solar corona using data obtained with the Visible Emission Line Coronagraph (VELC) payload onboard India's maiden solar mission, Aditya-L1.

The Aditya-L1 satellite generates Doppler velocity images of the solar corona, and these images are expected to provide information on the propagation of coronal mass ejections (CMEs) along the Sun-Earth direction, which has been unavailable to date.

Scientists at the Indian Institute of Astrophysics (IIA), who generated the Doppler velocity images, said it is well known that solar activity leads to the ejection of hot plasma. They said that the massive


ISRO's Solar Mission Aditya-L1 entering halo orbit.

and energetic plasma cloud ejected during powerful eruptions like the CMEs can travel either in the direction of the Earth or in any other direction from the Sun.

Scientists said that because CMEs can disrupt near-Earth space, it is important to infer their direction of propagation from the Sun as they erupt. "Universally, it is taught in high schools that the Doppler effect in sound helps to understand whether a train is approaching the railway station platform or moving away from it. A person standing on the platform would hear a steady increase in the pitch of the sound waves due to the train's whistle or horn when the train is coming into the station. The pitch would steadily decrease when the train is moving away from the station," scientists said.

Doppler velocity

Using a similar Doppler effect concept in light waves, the scientists have now generated Doppler velocity images.

"The positive and negative Doppler velocities indicate solar plasma moving away from and towards the direction of the observer, respectively," they said.

The detailed software for processing the VELC data is developed by Dr V. Muthupriyal, Senior Project Scientist with the VELC payload operations team in IIA, and her team. The technical details related to the data processing have been published recently in the Astrophysical Journal after international peer review.

According to R. Ramesh, Senior Professor in IIA and Principal Investigator, VELC project, the Doppler velocity images of the solar corona are unique. Only VELC has the capability to generate such images at present due to its one-of-a-kind spectroscopic observing capability in the visible wavelength range, he said. He added that the related processed images will be disseminated to the scientific community .

Why Big Tech is giving away open-weight AI models and what the companies want in return

From Muse Spark Open to Beam, Nemotron and Timestral, tech firms are releasing AI models to attract developers, challenge Chinese rivals and strengthen their ecosystems by attracting developers, to shape how AI is used and earn money from businesses, services and computing infrastructure built around their models; for Indian firms, these models are an alternative to relying on foreign AI

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CHENNAI

Beam. Muse Spark Open. Nemotron 3.5 Super VL. Nemotron 3.5 Bolt 74B-A9B. Timestral. These may sound like characters from a science-fiction franchise. They are, in fact, names of open-weight AI models announced or released by technology firms in U.S. and Europe in October.

Reflection AI released Beam's weights on October 16. Meta released Muse Spark Open while NVIDIA made two Nemotron models available on October 15. France's Mistral AI also announced Timestral, a family of models designed for time-series forecasting.

Firms are making models available for various tasks viz. from writing software and operating AI agents to interpreting images and forecasting time-series data. But why spend heavily to develop AI models and then allow others to download and adapt them? If the models are free to download, where does money come from?

The answer lies in a race to attract developers, shape how AI is used and earn money from the businesses, services and computing infrastructure built around these models.

Open-weight model

AI models learn patterns from data during training. Their trained parameters, called weights, are numerical values that help fix how the models process inputs and generate responses.

Think of an AI model as a chef who learnt from thousands of recipes. During training, the model learns patterns that help it decide what ingredients to combine and in what proportions to make a certain dish. Its weights shape these decisions. Making the weights available is like sharing the chef's acquired knowledge with other chefs, who can then adapt it to their own needs and create dishes of their own.

Most people access proprietary AI models via websites or apps while developers may access them via application programming interfaces (APIs). The provider controls access and may charge for usage.

An open-weight model makes its trained parameters available for download. Developers can run it on compatible computers or servers, adapt it for particular tasks or integrate it into applications.

This can give businesses and individual users greater control over how they deploy and customise AI models, and reduce dependence on a single provider. However, open-weight does not automatically mean open-source in the fullest sense. A firm may release weights without publishing training code or sufficient information about the data used to train the model. Licences may also curb some uses.

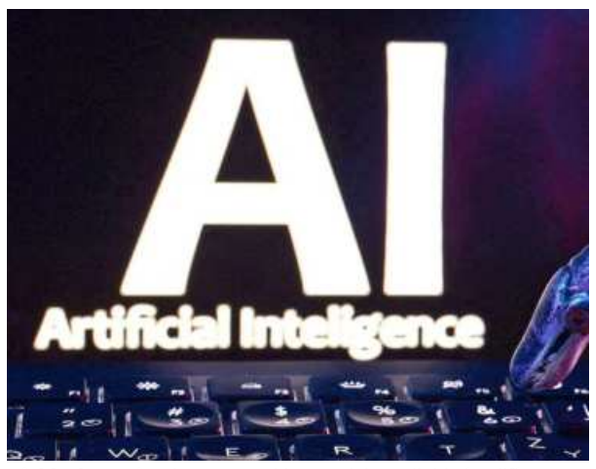
Open-weight AI is not new. OpenAI released GPT-2's weights in 2019. What has changed is the capability available via downloadable models and the emergence of Chinese developers such as DeepSeek as serious competitors, prompting a broader contest over who will shape the AI ecosystem.

Chinese competition

The growing availability of Chinese models has intensified competition in the open-weight AI market. Firms, including DeepSeek and Alibaba, released models whose weights developers can download and adapt. DeepSeek-R1, released in January 2025 under MIT licence, made a reasoning model available for others to use, modify and commercialise. Alibaba's Qwen family includes models of different sizes for a range of applications.

Developers choose models based on performance, cost, hardware requirements and licensing. If a Chinese model meets their needs at a lower overall cost, they may choose it over a proprietary American or European service.

As more developers build applications and tools around a model, its ecosystem can make it more attractive to others.



Highly dynamic: Firms are making models available for different tasks, from writing software to interpreting images. REUTERS

OpenAI made the geopolitical argument explicit when it released its open-weight models in August 2025. It said the models could help partner countries build AI infrastructure on what it called "U.S.-led rails", giving them an incentive to choose "democratic AI" over "autocratic AI". This is the company's stated position but it illustrates how the spread of AI technology is also viewed as a strategic contest.

Where is the money?

The answer depends on what the firm sells. Meta CEO Mark Zuckerberg outlined his approach in a July 2024 essay, Open Source AI Is the Path Forward. Drawing a comparison with Linux, he argued releasing Llama's weights may encourage firms to develop tools, integration and hardware optimisation around the models, aiding a broader ecosystem.

Mr. Zuckerberg said selling access to AI models was not Meta's business model, which centred on building products and services. He argued releasing Llama would help Meta avoid dependence on competitors' proprietary technology while encouraging the development of a broader ecosystem of tools and services around its models.

For NVIDIA, the commercial interest is different. It sells computing hardware and related software. Wider adoption of AI applications could raise demand for computing power, potentially benefiting NVIDIA, although developers can run open models on competing hardware. Revenue gains from individual model releases are not guaranteed.

Model developers such as Mistral can offer paid hosting, enterprise support, customised models and private deployments. Alibaba's Qwen models sit alongside its cloud and AI services. A developer may download a model's weights at no cost but pay a provider to host and run it reliably at scale. Alternatively, the developer may run it independently and never pay its creator. The business bet is that wider adoption will generate enough revenue or strategic value elsewhere to justify the cost of developing and releasing the model.

Shaping AI ecosystem

The longer-term prize is influence. A widely-used model can attract developers who build applications and tools around it. These additions can make the model easier to deploy and use, potentially attracting more developers. This reinforcing cycle is known as a network effect.

Distribution matters too. Making model weights downloadable is only one way to encourage adoption. Offering access via APIs, model hubs and inference platforms can make models easier for developers to discover, test and integrate into applications.

Mr. Zuckerberg argued Meta wants Llama to turn into an industry standard. In a 2024 submission to the U.S. National Telecommunications and Information Administration, OpenAI identified brand recognition, recruitment and attracting developers as reasons firms release model weights. It also warned that widely distributing weights can make misuse harder to control, since users can modify models and remove safeguards.

The strategy is therefore not simply about giving technology away. It is about encouraging developers to build on a firm's models, potentially raising demand for its infrastructure, tools and services as AI applications evolve.

There are risks too

Not everyone believes more openness is always better. Anthropic CEO Dario Amodei argued open-weight models can be a public good but warned releasing models with dangerous capabilities can make harmful uses harder to prevent. Once weights are widely distributed, they cannot be withdrawn.

OpenAI has made a similar argument saying that access through an API allows a provider to monitor use and restrict access, while downloaded weights can be run independently, beyond the original developer's control. The trade-off is that open weights can widen access and encourage competition, but reduce the original developer's ability to intervene when a model is misused.

What it means for India

For Indian businesses, open-weight models offer an alternative to relying exclusively on foreign AI services.

Organisations can adapt models to Indian languages and local requirements and may choose to run them on their own infrastructure, depending on their computing resources and the model's licence.

India has its own efforts too. Sarvam AI has released models, including Sarvam 30B and Sarvam 105B while AI4Bharat at IIT Madras has made Indian-language models and tools available to developers.

BharatGen has also released models for Indian languages and domain-specific applications, including agriculture, law and Ayurveda. The objectives, however, vary. Commercial firms may seek revenue via applications and enterprise services while research initiatives may prioritise language access and domestic technical capability.

Open weights alone do not guarantee lower costs. Hardware, maintenance, performance and licensing still matter and organisations must assess if running a model themselves is more economical than using a hosted service.

India-New Zealand free trade agreement takes effect

The India-New Zealand Free Trade Agreement (FTA), signed on April 27, came into effect on Tuesday, providing duty-free access to a range of Indian goods, including products from labour-intensive sectors such as textiles and leather. It is the seventh trade pact to come into force in the last five years. The others are agreements with the UAE, Mauritius, the European Free Trade Association (EFTA) bloc, the U.K., Oman and Australia. Signed on April 27, the agreement follows negotiations launched in 2010, stalled in 2015 and revived in March 2025. The pact becomes India's 17th operational FTA. New Zealand Prime Minister Christopher Luxon said "As of today, our historic Free Trade Agreement with India is in effect. Working closely with Prime Minister @narendramodi, we've delivered a deal that will help grow both of our economies". He said from Tuesday, Kiwi businesses can reach over a billion more clients with significantly cut tariffs. New Zealand has committed $20 billion FDI in over 15 years. PTI

Scientists unearth new instance of ‘contagious cancer’ in animals

Researchers studied melanoma in the brown bullhead catfish, a resident of Lake Memphremagog, between Quebec in Canada and Vermont in the U.S.; the finding is interesting because it could reveal clues to keeping a human’s body from rejecting a transplanted organ

D.P. Kasbekar

Ordinarily, cancers are not regarded as contagious – whereas AIDS, COVID-19, tuberculosis, and malaria are. Contagious diseases spread from one person to another through direct contact, airborne droplets or through insect vectors (like mosquitoes). On the other hand, cancers originate from one of the patient’s own cells.

However, so far, scientists have documented three instances in which naturally occurring cancers in animals became contagious. That is, the cancer cells originated in one individual, then moved into another, formed tumours in its new host, and then moved on in search of still other hosts.

In 2006, scientists who reported a transmissible venereal tumour that spreads between dogs wrote that “the concept of a contagious cancer cell has tended to be greeted with scepticism by many oncologists and immunologists”.

Interest among scientists

But recently, researchers from the University of Vermont in the U.S. reported the fourth instance. They studied melanoma in the brown bullhead catfish (*Ameiurus nebulosus*), a found in of Lake Memphremagog, a 51-km-long lake straddling the border between Quebec in Canada and Vermont in the U.S.

Melanomas are aggressive cancers that originate from skin cells called melanocytes. Melanocytes make the pigment melanin, which gives skin its colour. About 30% of catfish collected from across the lake since 2013 were found to be afflicted with melanoma.

The team’s findings were published in *Nature*.

Scientists are also interested because the work could reveal clues to keeping a person’s body from rejecting a transplanted organ. This is because the affected catfish’s immune system does not reject the cancer cells despite the latter being obviously non-self.

Mitochondrial DNA

The genome of cancer cells is often unstable. The DNA accumulates many changes in its sequence, allowing cells to grow aggressively and evade the body’s immune surveillance.

The Vermont researchers compared the genomes of the melanomas in the catfish with that of the normal cells from individual affected fish. They were surprised to find evidence that the melanomas had originated in a distantly related fish and then had invaded, parasitised, and formed tumours in new hosts. The researchers compared the mitochondrial DNA (mtDNA) from melanomas with the mtDNA from non-cancerous cells. Mitochondria, the cells’ powerhouses, are the descendants of a bacterium that an ancient single-celled organism engulfed, and evolved over billions of years to give rise to animal and plant life.

Over these aeons, many of the bacterium’s genes were transferred to the host cell’s nucleus, leaving behind only a small rump of mtDNA. In humans, the mtDNA content is five-million-times smaller than the DNA in the cell’s nucleus.

Tumour cells

The researchers found that the mtDNA in the cancer cells contained almost an order of magnitude more changes compared to the mtDNA in normal cells – a difference that natural variations can not explain. More analysis of the mutations indicated that the affected catfish had acquired their cancer cells from ‘outside’, which then formed tumours in their new host.

The team also found that the cancer cells had originated from beyond Lake Memphremagog.

Reproduction likely played a role in infection. During spawning, the fish come together in relatively small areas, and physical interactions amongst them could have given cancer cells the chance to move between individuals.

An alternative explanation for the high frequency of melanomas is a cancer-causing virus or microbe. The researchers screened all the tumour and normal tissues for viruses and other microbial agents but did not find any unique to affected fish.

Based on this they wrote: “although we cannot rule out a role for conventional pathogens in the original emergence or ecological transmission of this disease, the genomic evidence presented here overwhelmingly supports a transmissible cancer in which the tumour cells themselves act as the infectious entity”.

Other examples

Of the three previously documented instances of transmissible cancer, the one in dogs involved sex. In the 2006 paper, scientists found that the DNA from cancer cells of 40 affected dogs from across the world had mostly the same sequences, suggesting that the ‘infection’ had originated in one dog around 11,000 years ago.

The oral-facial cancer in Tasmanian devils, an endangered Australian marsupial, obstructed the animal’s ability to feed, thus leading to these animals’ population in the wild crashing by 90% and threatening to render them extinct. In all the affected Tasmanian devils, the tumour cells contained the same complex rearrangement of chromosomes, once again indicating a common origin, from a single cell, which then multiplied and spread across individuals, probably as they bit one another. Scientists have also found transmissible cancers in several bivalve molluscs, including clams, mussels, and cockles. Their cancer cells were probably transmitted passively from one individual to another via the seawater. In one clam species, scientists found the cancer had originated in a different species and then had jumped to this one, which is more unusual.

In fact, the authors of the latest study have speculated that the catfish cancer cells, like those of the bivalves, might be transmitted indirectly through the aquatic environment, and have become grafted to a host through its gills, skin or mouth.

‘Cooperation breakdown’

While the discovery of this fourth instance of a cancer that can move between bodies is not surprising, it also extends an unconventional understanding of the ‘disease’. In a complex biological organism, billions of cells cooperate to keep the organism alive and kicking.

Cancer arises when some cells refuse to cooperate, and proliferate at the expense of other cells. A transmissible cancer, however, implies that the cells break the rules in their original body as well as can exploit entirely different organisms.

In a 2015 paper, scientists wrote that the possibility lends weight to the idea that cancer is not just the product of specific mutations but “a breakdown of multicellular cooperation”.

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Melanoma mystery: A new study suggests that catfish cancer cells might be transmitted indirectly through the aquatic environment, and may have become grafted to new hosts through their gills, skin or mouth. WILL TURNER/UNSPLASH

THEGIST

Researchers studying brown bullhead catfish in Lake Memphremagog found that about 30% carried melanoma, a skin cancer that appears to spread between fish as living cancer cells rather than arising independently in each animal

The team found that tumour cells had far more mitochondrial DNA mutations than normal cells, indicating the cancer cells were acquired from other fish and later formed tumours in new hosts; evidence suggests the cancer originated beyond Lake Memphremagog

Scientists believe the transmissible cancer could help explain how foreign cells evade immune rejection; the affected catfish do not reject the non-self cancer cells, a phenomenon that may offer insights relevant to organ transplantation and immune response

BIG SHOT



Journalists film the exploration and return modules of the Martian Moons eXploration (MMX) spacecraft at the Tanegashima Space Center in Japan in August. Japan launched the MMX on October 20, en route to potentially becoming the first mission to retrieve samples from one of Mars’s moons. AFP

WHAT IS IT?

Heat storms: giant heat waves

Reuters

Heat waves have evolved from sporadic, isolated episodes to massive, cascading weather events that can blanket vast swaths of the planet for months or even years at a time.

To continue calling such sprawling phenomena ‘heat waves’ would be like calling a monsoon a ‘rain shower’. To understand the new reality of these immense expanses of heat, a consultation with two dozen climate researchers and an analysis of temperatures worldwide for for every hour since 1960 has yielded a new concept: heat storm.

Scientists and researchers often discuss global warming in the context of risks that will arise if the planet’s average global temperature rises more than 2 degree celsius above pre-industrial levels. An ordinary person can’t make much of that. However, heat waves are the connective tissue between the science of global warming and our everyday experience. And today supersized heat waves regularly swarm the planet.

Experts have also identified a ‘heat cascade’ as heat storms that run one after the other, simultaneously or both. Their duration has also swelled from around two months in the 1960s to nine months in the early 2000s.

Naming events is “a powerful way for people to understand that something is happening,” Anthony Leiserowitz, director of the Yale Program on Climate Change Communication, said. Yet “we don’t do that with heat waves,” he said, even though heat has been estimated to kill almost half a million people a year.

The largest heat storm on record, named ‘Vera’, lasted for 367 days over 2023-2024 and swept over nearly 85% of the globe. The world is now in the grip of a heat storm named Raimundo: it broiled Europe and stoked massive wildfires in Spain and France this summer. Since its inception on February 28, 2026, Raimundo has touched almost three-quarters of the planet’s surface.



People sit in the shade on the Spanish Steps during a heatwave in Rome, Italy. REUTERS